

38. Explain the different kinds of solid wastes. (B)
[Ans. 3:13:2]
39. How could solid waste pollution be controlled? (B)
(or)
How could solid wastes be disposed off? (B)
[Ans. 3:13:4]
40. Explain the methods of managing the garbage problem. (B) [Ans. 3:13:5]
41. Discuss the different types of solid wastes, the methods of disposing them and manage the solid waste problem. (A) [Ans. 3:13:2 + 3:13:4 + 3:13:5]
42. Discuss the role and responsibilities of individuals in controlling air pollution. (B) [Ans. 3:13:6]
43. Define 'Environmental Hazard'. (C) [Ans. 3:14:1]
44. Classify the important natural disasters. (C)
[Ans. 3:14:3]
45. Write a short note on each of the following :
 - i) Earthquake (C) [Ans. 3:14:3:01]
 - ii) Landslide (C) [Ans. 3:14:3:03]
 - iii) Tsunami (C) [Ans. 3:14:3:02]
 - iv) Flood (C) [Ans. 3:14:3:05]
 - v) Cyclone (C) [Ans. 3:14:3:06]
 - vi) Volcanic Explosion (C) [Ans. 3:14:3:04]
46. Discuss briefly the different types of natural disasters and the methods of managing them. (A)
[Ans. 3:14:3 + 3:14:3:01 to the end of 3:14:3:06, all in full]

Chapter - IV

ENVIRONMENTAL PROBLEMS, POLICIES AND PROTECTION OF ENVIRONMENT

4:00 Introduction

Just like the culture, the language and the life patterns of people, the environment of India also is diverse with beautiful and bountiful forests, mighty rivers, snow-bound peaks, deep seas, rich mines and a variety of minerals, vast agricultural and pasture lands, live stock, wild life and flora and fauna of countless varieties. It is a vast treasure home of natural resources.

Just as the natural resources of India are diverse and vast, the onslaught on them too are multiple and innumerable. From the Ganges of North to the Amaravathy and Noyyal rivers of Southern Tamilnadu, all the waterways in India are getting reduced to drains carrying sewage and industrial effluents. The Governments at the Centre and State are undertaking several measures to cleanse the major rivers and prevent any further damage to our environmental resources. This lesson attempts to study the various environmental issues and policies followed by the governments in India and also the role of NGOs in the protection of environment, especially

Environment Foundation of Africa, Worldwide Fund for Nature, Conservation International, Green Peace and International Union for Conservation of Nature.

4:01 Major Environmental Problems

4:01:1 Global Warming

The level of carbon dioxide and carbon monoxide in the atmosphere is going on increasing, due to which the average temperature in all the regions across the world has raised significantly. This is referred to, as '**Global Warming**'. It is said that the average temperature on the earth has raised about 3°C during the last 30 years. As a result of global warming, the following effects have been observed.

- i) Glaciers have started melting fast, flooding the rivers which results in the rise of sea level, immersing the adjoining low lying areas.
- ii) Seasons in many parts of the world become erratic.
- iii) There may occur severe famine for drinking water across the world.
- iv) Severe water shortage will lead to significant reduction in agricultural output which may trigger widely poverty and food shortage in the world.

- v) Due to growing power needs, power cut will be imposed for long hours in a day, crippling the functioning of industries.
- vi) Various kinds of diseases (like swine flue) will spread fast on global level.
- vii) Due to sea-erosion, land area will shrink.
- viii) As desertification spreads, soil fertility is adversely affected leading to drought in many parts of the world.
- ix) As agriculture gets crippled, food shortage will be found globally.

4:01:2 Green House Effect

The atmosphere on the earth has been divided into many layers based on the height and temperature. Among these the 'Troposphere' at the bottom and the 'Stratosphere' at top are the most significant ones. In the equatorial region troposphere extends upto 17 K.M. in height from earth surface and in polar regions it is upto 8 K.M. Troposphere contains Nitrogen, Oxygen, Carbon dioxide, Water vapour and other gases. The amount of CO_2 and water vapour contained in the atmosphere, is responsible for stabilising the atmospheric temperature.

When the level of Carbon dioxide in the atmospheric air increases, it leads to the phenomenon

of '**Greenhouse effect**' by which global temperature rises significantly.

Greenhouse effect is an analogy to the heat-trapping process caused by the gases (particularly Carbon dioxide) present in the atmosphere, 20 to 50 km. above the earth's surface. Greenhouse is a building with transparent walls and roof made up of glass, used for the cultivation of plants under controlled conditions of light and temperature. In a greenhouse, the visible light of high frequency (short wave length) from the sun, passes through the glass walls and roof and warms up the soil. The warm soil radiates the heat energy in the form of low frequency (longer wavelength) infra-red rays (IR). Glass is opaque (i.e. does not permit to pass through) to infra-red radiation. Hence the glass walls and roof of the greenhouse partly reflect and partly absorb the IR rays emitted by the soil surface. In that process, the outgoing IR rays are not allowed to escape the greenhouse and get into the space outside. The IR rays thus trapped inside, warm up the air inside the glasshouse. This mechanism keeps the greenhouse warmer than the surroundings. This "**heat-trapping**" process is called '**Greenhouse effect**'.

Just like the glasses of the Greenhouse, the transparent Carbon dioxide and water vapour present in the atmosphere allow the incoming visible light of

high frequencies to reach the earth's surface and warm it up. The warmed up earth's surface then emits heat energy in the form of low frequency IR rays which are prevented by the atmosphere layer containing high level of Carbon dioxide, to escape into the outer space. In fact the outgoing IR rays are reflected back to the earth's surface, due to which the temperature of the earth's atmosphere steadily rises up, which is known as '**Global-warming**'. If emission of greenhouse gases (gaseous constituents of the atmosphere like CO_2 , CH_4 , Oxides of Nitrogen, Chlorofluoro Carbon, etc. which absorb and re-emit infra-red radiation) continue, global warming will increase by 0.2°C to 0.5°C per decade.

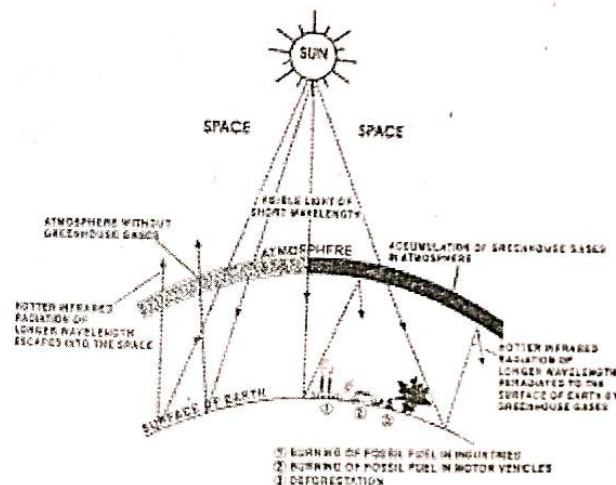


Fig. 4.1 Mechanism of greenhouse effect. In one half of the diagram, atmosphere is normal. In another half, atmospheric effect is shown.

4:01:3 Climate Change

The weather of a place goes on changing depending upon the pattern of wind flow, humidity in air, amount of rainfall and prevailing temperature. The average weather conditions of a place i.e. the maximum and minimum temperatures constitutes its climate. The climate in the world, in general had remained more or less stable for the past many centuries. But due to intense human activities, the global climate has changed a lot in the last fifty years or so, as contended by **"Inter-governmental Panel on Climate Change" (2003)**.

The Principal reason for climate change is the excessive use of fossil fuels such as coal and petroleum. These fossil fuels contain high amount of carbon, which on burning emit carbon dioxide (CO_2). Due to increasing use of fossil fuels, the level of CO_2 in the atmosphere has gone up by 30% in the last 50 years. This has resulted in increasing the global temperature between 10°F to 20°F .

If the present rate of consumption of fossil fuel continues for long, then there will be significant increase in global temperature which may result in increasing the level of the seas, melting of glaciers

in the Arctic and Antarctic regions, flooding of rivers followed by drought, drastic changes in the amount of rainfall and snowfall, longer summer with high temperatures, and shorter winter with very low temperatures. The seasons in the climate too get changed.

The drastic climatic changes will result in the emergence of environmental hazards to human health such as extreme weather, ozone depletion, loss of biodiversity, stress to food producing systems and the global infectious diseases. The world Health Organization (WHO) estimated that 1,60,000 deaths, since 1950 are directly attributable to climate change. Many believe this a conservative estimate. **Global Humanitarian Forum (2009)**, estimated more than 3,00,000 deaths and about \$125 billion in economic losses each year, and indicating that most climate change induced mortality is due to worsening flood, and droughts in developing countries.

4:01:4 Ozone Layer Depletion

The stratosphere which is at a height of 72 K.M. from earth's surface, contains scarce amount of water vapour. A thin layer of ozone gas, less than 1 ppm is also present, which absorbs most of the ultra violet (UV) rays. The ozone layer is very thin protective

layer in the stratosphere. Though constituting less than 1 ppm of the gases in the atmosphere, it absorbs most of the ultra-violet (UV) rays from the sun. The UV rays influence the body in many ways. Out of its beneficial effects is the formation of vitamin D in the

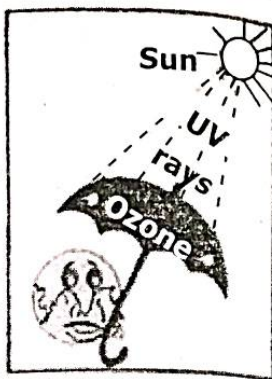


Fig. 4.2 Ozone Layer

skin. UV radiation also causes ageing of the skin and skin cancer. It influences the immune system by suppressing the immune defence against tumours. It is estimated that with a decrease in 1 per cent of ozone, there will be 3 per cent increase in non-melanoma skin cancer. UV rays may also increase the incidence of diseases like measles, chickenpox and other viral diseases affecting the skin by inducing rash, bacterial infections like tuberculosis and leprosy and fungal infections. UV radiation has also been found to damage the eye by the formation of cataracts. A study by the US Environment Protection Agency estimates a 0.3 to 0.6 per cent increase in the incidence of cataracts by the decrease of every 1 per cent of the stratospheric ozone. Plants are also found to be affected during photosynthesis.

From the foregoing discussion, one can realise the importance of the ozone layer from the health point of view. In 1985, atmospheric scientists of the British Antarctic Survey came out with a startling report indicating the vast depletion of ozone in the atmosphere over the Halley Bay in Antarctica. According to the report, between 1977 and 1984 more than 40 per cent of the ozone layer had decreased. It was found that the ozone depletion or the ozone hole was wider than the continent and extended about 12-24 km in altitude. The concentration of ozone is decreasing at a rate of 0.3 per cent every year. At this rate all life on earth may come to an end within 60 years. A similar hole was also noticed over the thickly populated northern hemisphere. In 1988, under the Montreal Protocol, developed countries agreed to eliminate 90 per cent of the ozone depleting substances by 2000. All the nations in the World are taking concerted efforts to minimize the impact and save the earth from any further damage.

4:01:4:01 Causes of Ozone Depletion

It has been found that the Chemical Chlorofluoro Carbon (CFC) is one of the main destroyers of the ozone layer. The chlorofluoro carbons are made up of

chlorine, fluorine and carbon. When the CFC was first introduced in the 1930s, it was called a miracle gas because of its non-poisonous, non-inflammable and highly stable nature. It was soon put to a variety of uses like coolants in refrigerators and air-conditioners, as propellants in aerosol sprays, in blowing plastic foams and as cleansers for electronic parts in computer circuits. The chlorine atom in the CFC disturbs the equilibrium in the formation and destruction of ozone. Chlorine acts like a catalyst and it does not get destroyed in the process. One CFC molecule can destroy 1,00,000 molecules of ozone. The CFC released into the atmosphere finds its way into the stratosphere and remains there.

When the chlorine atom in the CFC collides with an ozone molecule, it splits up the ozone molecule into oxygen, forming chlorine monoxide (ClO). When this chlorine monoxide radical collides with an oxygen atom, it forms an oxygen molecule and a free chlorine is given out. This free chlorine collides with a fresh ozone molecule and the reaction continues.

It is estimated that 100 grammes of CFC can destroy about 3.5 tonnes of ozone. The approximate annual release of CFC is 1.1 million tonnes.

4:01:5 Measures to Control Impact on Atmosphere

- i) Controlling the level of CO_2 in the atmosphere by raising large number of trees possible, apart from curtailing deforestation.
- ii) Drastically cut the use of fossil fuels in our day to day life.
- iii) Using bio-insecticides instead of synthetic pesticides.
- iv) Developing new technologies, replacing the ones which use chloro-Fluro-Carbon (CFC) as the refrigerant, causing heavy damage to the ozone layer.
- v) Developing technologies to recycle waste water from industries that could be used to raise plants and trees.

4:01:6 Acid Rain

Acid rain is a kind of chemical leprosy that is eating into the face of North America and Europe. We cannot see acid rain but we can see its signature, an indicator in the haze patterns that it leaves on monuments and forests. Acid rain, generally known by the term 'acid precipitation', reaches the earth in the form of acid snow, acid sleet, acid fog, acid frost, acid dew as well as acid rain,

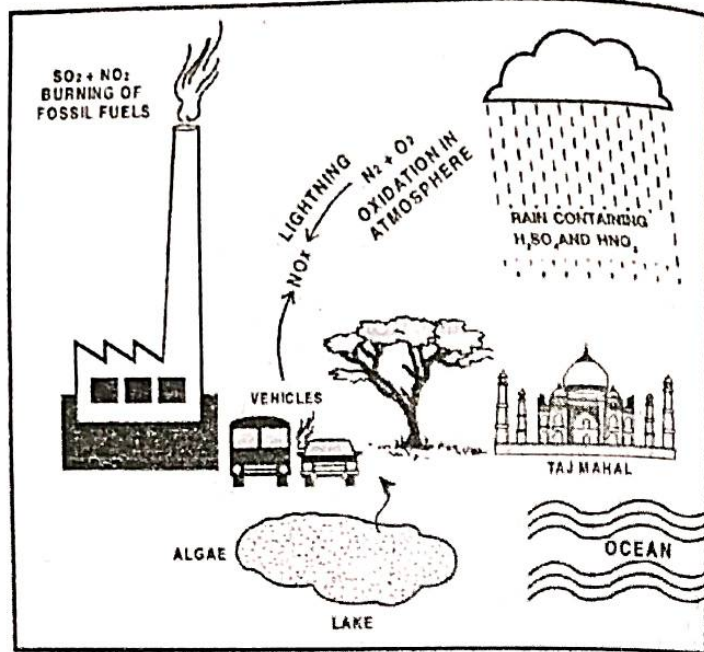


Fig. 4:3 Formation of acid rainfall and its impact on Taj Mahal

Acid precipitation is caused by the emission of sulphur dioxide and nitrogen oxide from the combustion of fossil fuels. Two-thirds of SO_2 comes from coal-fired and oil-fired power plants. SO_2 does not remain in the gaseous state for a long time; very soon it reaches and reacts with moisture to form sulphuric acid. Similarly, nitrogen oxides released into the atmosphere from the automobile exhausts are transformed into nitric acid. Very often the air currents carry them thousands of miles from the

source. What started as a problem on a small scale as acid precipitation has assumed gigantic proportions with the advent of the industrial age.

Acid rain has an impact on the biosystem. Today lakes in more than 13000 square kilometres of north Norway are practically devoid of fish. Researches are going on in Norway, Sweden, United States of America and Canada on the effect of acid rain. At pH value 5.5, Trout and Salmon populations are eliminated. Below pH 4.8, lakes are generally fishless. In addition to loss of fish life, loss of hundreds of other organisms such as certain types of algae, crustacea, molluscs and insects is also reported. Acid rain reduces forest growth and increases the amount of calcium and other nutrients leached from agricultural soil. Nearer home it is reported that the white marble with which the treasure of love Taj Mahal was constructed, is now turning into yellow because of the deadly brew of gases and matter, mixing with moisture in the air which comes down as acid rain.

4:01:7 Extinction of Flora and Fauna

At the Earth Summit in June 1992, the majority of the world's nations signed a convention of Biological Diversity to conserve whatever remains of

the earth's astounding plant and animal diversity. This global agreement had taken shape against the backdrop of awesome destruction of the Earth's flora and fauna. Of the 1.5 million species known to inhabit the Earth (humans are just one of them), an estimated one-third to one-fourth are threatened with extinction within the next few decades.

Because of the extensive habitat changes brought by mankind, we are losing a species a day.

Man's greed is destroying rainforests, the home of half of the world's life forms. While one may question the right of man to ride roughshod over the rest of the creation, what is more important is the impact that this loss of biodiversity has on human survival itself. The majority of the world's population still depends on wild plants and animals for their daily food, housing, medicine, fodder, fuel wood, and spiritual sustenance. For these billions of people, the loss of biodiversity is a direct attack on their livelihood and social security.

Also, traditional farmers the world over, have developed a variety of crops and livestock. This precious resource too has been eroded over the last few decades with lakhs of traditional crops and

livestock breeds being replaced by cash crops or laboratory generated hybrids.

India's biological diversity is one of the richest in the world. As many as 45,000 species of wild plants and over 77,000 species of wild animals have so far been recorded in the country. Apart from the 4,000 species of wild plants, its living resources included 370 mammals, 1200 birds, 180 amphibians, 1700 fishes, and 400 reptiles. This comprises about 6.5 per cent of the world's known wildlife. As many of the biologically rich areas in the country are yet to be exploited scientifically, many more species may be discovered in India in the not so distant future. At least 166 species of crops and 320 species of wild relatives of crops are known to have originated in our country. Within each of these species, the variety is astounding. For example, there is an estimated 50,000 to 60,000 varieties of rice grown in India till sometime back. Similarly significant is the indigenous livestock diversity with 27 breeds of cattle, 40 sheep and 22 goats.

Unfortunately, 10 per cent of India's recorded wild flora and fauna is on the verge of extinction. In the last few decades, it has lost at least half of its

forests, polluted over 70 percent of the water bodies, besides degrading most of the coasts. While no one is sure how many species have been lost already, the cheetah (*Acinonyx jubatus*) and the pink-headed duck (*Rhodonessa caryophyllacea*) are among the few conspicuous ones that are no more. Similarly thousands of crop strains have disappeared and not all of these remain even in the gene banks. Since the advent of the Green Revolution, a handful of seeds generated in the laboratory have replaced the innumerable countless Indian varieties.

Sadly, in countries like India, the development policies and projects have hardly been sensitive to the need for biodiversity conservation, and that of the local communities. The government's inability to remove poverty and curb the consumerism of middle classes resulted in conditions in which sensible natural resource management has been accorded low priority. Therefore, the Indian government has to give a serious thought to the prevailing development policies. Article 6 of the Biodiversity Convention commits countries to "integrate, as far as possible and as appropriate, the conservation and sustainable use of biological diversity into relevant sectorial or cross-sectorial plans, programmes and policies."

Article 14 of this Convention requires that the countries, "introduce appropriate procedures requiring Environmental Impact Assessment of its proposed projects that are likely to have significant adverse effects on biological diversity with a view to avoiding or minimizing such effects".

4:01:7:01 Factors Responsible for Flora and Fauna Extinction

- Water Pollution
- Air Pollution
- Acid Rain
- Global Warming and Ozone Depletion
- Population Explosion
- Industrialization
- Urbanization
- Deforestation
- Thermal and Nuclear Power Plants
- Interruption of the food chain and the resultant food scarcity
- Floods, earthquakes and volcanic eruptions
- Indiscriminate use of chemical fertilizers and pesticides
- Shortage of rainfall

4:01:7:02 Measures to Conserve Flora and Fauna

- The government should revise the agricultural policy and incorporate conservation and upgradation of indigenous crop and livestock varieties.
- Biodiversity concerns should be incorporated in the existing Environmental Impact Assessment procedures.
- A fresh look should be taken at the conservation policies, programmes and laws in keeping with Article 6 of the Biodiversity Convention, which requires nations to "develop national strategies, plans or programmes for the conservation and sustainable use of biological diversity or adapt for this purpose existing strategies, plans or programmes."
- Lopsided aspects of the current conservation programmes should be reviewed. For instance, the government should not confine conservation to some of the glamorous animals, mammals and birds, alone. It should also concentrate its efforts on the rest of the biological diversity too.
- Local communities should be involved in the conservation efforts.

The convention on Biodiversity (Earth Summit) stipulated the following measures for conservation of biological diversity:

Each contracting party shall as far as possible and as appropriate:

- Establish a system of protected areas where special measures need to be taken to conserve biological diversity.
- Develop, where necessary, guidelines for the selection, establishment and management of protected areas, or areas where special measures need to be taken to conserve biological diversity.
- Regulate or manage biological resources important for the conservation of biological diversity whether within or outside protected areas with a view to ensuring their conservation and sustainable use.
- Promote the protection of ecosystems, natural habitats and the maintenance of viable populations of species in natural surroundings.
- Promote environmentally sound and sustainable development in areas adjacent to protected areas with a view to furthering protection of these areas.

- Rehabilitate and restore degraded ecosystems and promote the recovery of threatened species inter alia, through the development and implementation of plans or other management strategies.
- Establish or maintain means to regulate, manage or control the risks associated with the use and release of living modified organism resulting from biotechnology which are likely to have adverse environmental impacts that could affect the conservation and sustainable use of biological diversity, taking also into account the risks to human health.
- Prevent the introduction of control or eradicate those alien species, which threaten ecosystems, habitats or species.
- Endeavour to provide the conditions needed for compatibility between present users and the conservation of biological diversity and the sustainable use of its components.
- Subject to its national legislation, respect, preserve and maintain knowledge, innovations and practices of indigenous and local communities embodying traditional lifestyles relevant for the conservation and sustainable use

- of biological diversity and promote their wider application with the approval and involvement of the holders of such knowledge, innovations and practices and encourage the equitable sharing of the benefits arising from the utilization of such knowledge, innovations and practices.
- Develop or maintain necessary legislation and/or other regulatory provisions for the protection of threatened species and populations.
 - Where a significant adverse effect on biological diversity has been determined pursuant to Article 7 (identifying and monitoring components of biological diversity important for conservation), regulate or manage the relevant processes and categories of activities.

4:02 Environment Protection and Policies in India

4:02:1 Conservation of Environment

As we already mentioned that earth is the only planet on which life is possible. Population on this earth is also increasing at an alarming rate and it is estimated that the population will increase upto about 818 crores or more in 2025, putting stress on natural resources.

Basically there are two main reasons for the exploitation of environment and its resources, one is poverty and other is the rich people. Both have over exploited the resources.

Although development is necessary for man but it should be sustainable and should not destroy the resources but conserve their development. Environment conservation means proper management. Available resources on this earth should be used in such a way that it should not be depleted. Environment management is possible only by the awareness among people.

4:02:2 Need for conservation

To safeguard the natural environment from the onslaught of continuous degradation as we cannot stop development, protection measures are very much necessary. Protection is that part of the resource management that is concerned with the discharge into the environment of substances that might be harmful and with safeguarding beneficial uses. Environmental pollution is said to be the gift of modern technological development, rapid growth of industrialisation and urbanisation. The environmental damage is now assuming a dangerous proportion throughout the world and a growing awareness is

discernible to maintain ecological balance for the future generations. Throughout the last quarter of the century a global concern to protect environment is witnessed to preserve our right to life. Though the subject is science, the environment problems are really social problems. They begin with people as the cause and end with people as victims. Though the protection of environment is not new to the Indian culture, as the environmental degradation is rapid in its growth, fast measures are to be taken to protect the environment.

4:02:3 Objectives of Conservation

Objectives for the conservation is to protect the present resources and situation so that good, clean and self-sufficient environment could be made available to the future generations also. The following are the objectives:

1. Proper ecological balance including biotic community management and other basic components like earth, air, soil, water, trees etc.
2. To keep the diversity of organisms on earth and their development.
3. For the conservation of natural resources for a long time on this earth **"Think globally and act locally"** – Rene Dubos.

4:02:4 Environmental Conservation Measures Taken in India

To protect the environment, Indian Government resorted to certain activities like laws, seminars, camps, amendments in the Constitution etc.

Lists of Acts for protection of Indian environment since 1847.

- 1847 Indian Fisheries Act
- 1905 Bengal Smoke Nuisance Act
- 1912 Bombay Smoke Nuisance Act
- 1917 Mysore Destruction by Insects and Pests Act
- 1919 The Poison Act
- 1919 Andhra Pradesh Agricultural, Pest and Disease Act
- 1923 The Indian Broilers Act
- 1927 The Indian Forest Act
- 1946 Bihar Wastelands Act
(Reclamation, Cultivation and Improvement)
- 1947 Mines and Minerals Act (Regulation and Development)
- 1948 The Factories Act (Pollution and Pesticides)
- 1949 Andhra Pradesh Improvements Scheme Act
(Land Utilisation)
- 1951 Industries Act (Development and Regulation)
- 1953 Orissa River Pollution and Prevention Act

- 1954 Assam Agricultural Pests and Disease Act
- 1954 Prevention of Food Adulteration Act
- 1955 Acquisition of Land for Flood Control and Prevention of Erosion Act
- 1956 River Boards Act
- 1958 Ancient Monuments and Archeological Sites and Remains Act
- 1958 Kerala Agricultural Pests and Disease Act
- 1962 Atomic Energy Act (Radiation Protection Rules 1971)
- 1963 Gujarat Smoke Nuisance Act
- 1968 The Insecticides Act
- 1969 Maharashtra Prevention of Water Pollution Act
- 1970 Merchant Shipping (Amendment) Act
(Harbour and Coastal Water Dumping of Oil etc.)
- 1972 Wild Life Protection, Act
- 1974 Water Act (Prevention and Control of Pollution)
- 1976 Urban Land Act (Ceiling and Regulation)
- 1977 Water Cess Act (Prevention and Control of Pollution)
- 1980 Forest Conservation Act
- 1981 Air Pollution Prevention and Control Act
- 1986 Environment (Protection) Act

- 1992 Policy Statement for Abatement of Pollution by Ministry of Environment and Forests.
- 2008 Hazardous Wastes (Management Handling and Transport) Rules
- 2010 The National Green Tribunal Act
- 2011 E-Waste (Management and Handling) Rules

4:03 Constitutional Amendments Made in India

To safeguard environmental conservation, the following amendments were made in the constitutional provisions:

42nd Amendment in the constitution: The Indian constitution was amended in 1976 by late Mrs. Indira Gandhi. According to this 42nd amendment, the problems regarding forests, wild life and environment were considered. Laws have been made and the fundamental duties imposed on every citizen of this country. Article 48-A of this section states that "the state shall endeavour to protect and improve the environment and to safeguard the forests and wild life of the country",

Article 51-A clause (g). It deals with the fundamental duties of the citizens and it states that, "It shall be the duty of every citizen of India to protect

and improve the natural environment including forests, lakes, rivers and wild life and to have compassion for living creatures".

Department of Environment and Forests (1972): In 1972 different departments and scientists collectively worked for the study of various environmental problems and their solutions. In January 1982 for the protection of Environment, a committee was formed which works for proper management and handling of the laws. As per the recommendations of this committee a full fledged department of Environment had been started. Later this department was converted into Ministry of Forests and Environment.

4:03:1 Factories Act, 1948 (Amended in 1987)

Section 12, of constitution says that it is the duty of every industrialist to properly treat and manage and dispose waste in accordance with the plans made by the state Government.

1. In Section 11 of the act, cleanliness of the factories is taken into account. It is the duty of the factory owner to check that factory is free from any effluent.

2. Measures for any emergency in the factory.
3. Any harmful effluent producing factory should be away from habitats of human beings.
4. Limited quantity of harmful waste should be emitted.
5. Proper maintenance and check of the machineries.

4:03:2 Environment Conservation Act, 1980 (Amended in 1988)

In 1894 a policy was made which was amended in 1952 and 1988. These amendments have now become an environment act. According to this law-

1. Natural forests on the earth should not be converted into other plantation without acceptance of the Government.
2. For the conversion of land exceeding 20 hectares the acceptance of Government is necessary.
3. If one converted forest area for development project, an equal amount of area is to be planted by him.
4. Proper check on illegal land capture and crop-rotation.
5. Encouragement to tribal community.
6. Forest planning and management should be stressed.

7. Afforestation near the hills and slopes of the hills.
8. No area should be deforested with the purpose of reforesting it.
9. The cutting of trees in the hills should not exceed 10 hectares whereas in plains it should not exceed 20 hectares.
10. Grazing should be checked.
11. Penalties for any disobedience of laws.

4:03:3 Forest (Conservation) Act, 1980 (Amended in 1988)

A forest policy was framed in 1894, revised in 1952 and then in 1988. Indian Forest Act was promulgated in 1927 with amendments in 1930, 1933 and 1948. Forest (Conservation) Act was an improvement over it.

1. Diversion of forest land for non-forest purposes (including cultivation of rubber plants, palms, medicinal plants, spices, tea, coffee etc.) or conversion of a reserved forest into non-reserved one cannot occur without the prior approval of Central Government.
2. Forest land, less than 20 hectare, can be used for nonforest purposes with the approval of State Government. For more than this land, the approval of Ministry of Environment and Forests

will have to be taken which will set up an advisory committee for this purpose.

3. If diversion is allowed, compensatory afforestation must be undertaken on an equivalent area of non-forest land or double the area of degraded forest.
4. A control should be exerted over shifting cultivation and encroachments.
5. Tribal rights and concessions must be highlighted along with control mechanisms.
6. All forest working plans should stress conservation and have multidisciplinary approach.
7. As per the previous Forest Act (1927), the State Government has the power to declare any forest or waste land as reserve forests where grazing and felling are not allowed. Grazing, clearing for cultivation, erosion, maintenance of ground water and lines of communication are prohibited in reserve forests.
8. All critical areas in hills, catchment areas, steep slopes and other parts under erosion and landslides must be protected and quickly afforested. No area above 1000 m altitude should be opened for tree felling, unless and until it has fully recovered.

9. No forest area should be cleared of self grown trees and shrubs for the purpose of reforestation.
10. Clear felling blocks should not exceed a hectare in hills and 25 hectares in plains. Regeneration should be undertaken immediately.
11. Grazing problems of the area should be studied and appropriate measures adopted.

Six regional offices have been opened to monitor conditions of forests and observe steps required for their conservation. They are at Chandigarh, Lucknow, Shillong, Bhubaneswar, Bhopal and Bangalore.

Advantages :

- (i) Guidelines were issued in June 1990 to involve tribals, forest dwellers and village communities for regeneration of degraded forests on the basis of sharing.
- (ii) The various measures undertaken under the law as well as thorough policy guidelines have reduced the trend of deforestation from 1.3 million hectares per annum down to 0.15 million hectares and then to only 3600 hectares.
- (iii) With the coming up of forest protection committees, more than 0.5 million hectares of degraded land has come under forest cover.

Limitations :

- (a) As the timber prices rise, cases of excessive lumbering, stealing and smuggling of timber have increased.
- (b) Small, slow encroachments by individuals are on the rise.
- (c) There is an increase in the number of grazing animals. They stray into prohibited areas too often.
- (d) Fuel collectors often scrap and peel off the bark of trees.
- (e) Trees have to be lopped time and again to protect telephone and electricity supply lines.
- (f) There is no way to stop forest destruction for building roads, water supply schemes, schools, dispensaries, forest based industries, etc.
- (g) It is very difficult in forests to apprehend the offenders. The law takes a long time to punish the offenders.

4:03:4 Wildlife (Protection) Act, 1972 (Amended in 1991)

Wildlife is living component of nature which has not been tamed by humans. It includes both plants and animals. Wildlife is essential for maintaining

ecological balance, preventing soil erosion, obtaining a number of economic products, potential source for breeding improved varieties, new fodder, drugs, beverages, etc. Attempts have been made previously to protect wildlife through various legislative measures like (a) Madras Wild Elephant Preservation Act 1873, (b) All India Elephant Preservation Act 1879 (c) Wild Birds and Animals Protection Act 1912 (d) Bengal Rhinoceros Preservation Act 1932 (e) Assam Rhinoceros Preservation Act 1954. A Central Board of Wildlife was established in 1952.

A comprehensive Wildlife (Protection) Act was passed in 1972 which was further amended in 1991. Under this act there is an Indian Board of Wildlife chaired by Prime Minister and a Wildlife Advisory Board for each state. Indian Wildlife Board has asked both Botanical Survey of India (B.S.I) and Zoological Survey of India (Z.S.I) to prepare lists of threatened species of plants and animals respectively.

A threatened species, is the one which is likely to become extinct if not provided protection. It is called endangered species, if its number has been reduced due to an environmental factor and rare, if its number is small due to its localisation in a small area. **International Union for Conservation of**

Nature and Natural Resources (IUCNNR), Margis (Switzerland) maintains a data book of threatened species. Similar red data books have been prepared for threatened species of Indian plants and animals.

Under the Act, certain biogeographical areas have been set apart for providing natural habitat, resources and protection to threatened species and preserve biological diversity. They are of three types - national parks, sanctuaries and biosphere reserves (i) **National Parks**. They are reserve areas maintained by Government where cultivation, grazing, forestry and habitat manipulation are not allowed. India has currently 75 national parks spread in about 1% of the geographical area. (ii) **Sanctuaries**. They are tracts of land with or without lakes where animals are not hunted. Other activities like tilling of land, collection of forest products, lumbering, etc. are not allowed. India has about 421 sanctuaries spread in 3.2% of geographical area. (iii) **Biosphere Reserves**. They are multipurpose protected portions of different ecosystems where wildlife, tribals, domesticated plants and animals are allowed to live in harmony. Out of the proposed 14 sites, India has built 8 biosphere reserves (e.g. Nanda Devi, Nilgiri). Each biosphere reserve has a **core zone** where no

human activity is allowed, a **buffer zone** where limited human activity is allowed, **manipulation zone** where balanced human activity is allowed and **restoration zone** where corrective steps are taken to restore degraded area.

A Central Zoo Authority has been established to oversee and coordinate the activities of some 200 zoos in the country. There is a director of Wildlife Preservation with four regional deputy directors (New Delhi, Mumbai, Kolkatta and Chennai).

1. Wildlife Protection Act prohibits hunting and trapping of wildlife.
2. Trading in wildlife products like skins and ivory is banned.
3. Human activity is not allowed in the core zone of biosphere reserves and national parks.
4. Wild animals cannot be shot down or killed even as a protective measure unless and until the same are declared man-eater or otherwise harmful to human beings by Wildlife Warden.
5. Uprooting, picking and sale of endangered or rare plants of scientific, ecological or medicinal importance are not allowed.
6. No new arm licence is to be normally issued within ten km of a sanctuary.

7. All weapons, vehicles and tools used for committing a wildlife offence shall become the Government property.

Liberation

- (i) Enforcement of wildlife Act has helped in increasing population of many endangered animals
- (ii) Ecological balance has been restored at several places where human activity is restricted under the Act.
- (iii) It has opened avenues for recreation and tourist attraction.

Limitations

- (a) Stress is more on conservation of one or two species in an area. A holistic approach is missing.
- (b) Human beings have been uprooted at a number of places to make room for wildlife
- (c) Less attention is being paid to endangered plant species.
- (d) Exploitation of plant species continues under the garb of traditional medicine as all the plants of medicinal importance are not cultivated
- (e) It takes a lot of time for the wildlife authorities to declare an animal man-eater as dangerous. By that time the animal unleashes a reign of terror

in the area of its operation, mauling or killing domesticated animals, children and even adult human beings.

- (f) Apprehension of offenders and legal procedure against them are cumbersome

4:03:5 Water (Prevention and Control of Pollution) Act, 1974 (Amended in 1988)

It was preceded by (i) Orissa River Pollution Act, 1953. (ii) River Boards Act, 1956 (iii) Maharashtra Prevention of Water Pollution Act, 1969 (iv) Merchant Shipping (Amendment) Act, 1970. Water Act was enacted under article 252 (1) of Constitution as a social welfare measure to prevent and control water pollution and maintain or restore wholesomeness of water. Water includes rivers, rivulets, springs, canals, channels, ground water and coastal/tidal water. Pollution under the Act includes (a) Contamination of water (b) Alterations in physical, chemical or biological properties of water. (c) Direct or indirect discharge of sewage, trade effluents or any other substance (liquid, solid, gaseous) into water. (d) The discharge is likely to cause nuisance or render such water harmful or injurious to public health or safety or to domestic, commercial, industrial, agricultural

or other uses or to the life and health of animals, plants or aquatic organisms.

Under the Act, authority is vested in Central and State Water (Pollution Control) Boards.

Central Board : All the appointments are made by Central Government. The board has a full time nominated chairman possessing specialised knowledge or experience in environmental protection. There are a maximum of five central government officials, a maximum of five state board members, a maximum of three nonofficials representing interest in agriculture trade, fishery and industry, two representatives of central government owned companies or corporations and a full time member-secretary qualified in public health engineering with experience of environmental protection methods. Central Board is a registered corporate body. Barring member secretary all others hold office for a term of three years but can be renominated for two extra terms.

State Board : All appointments are made by the State Government. There is a whole time or part-time chairman with specialised knowledge, practical or administrative experience, a maximum of five state government officials a maximum of five representatives from members of local authorities, a maximum of three nonofficials representing

agriculture, fishery trade or industry, two representatives of state government companies or corporations and a full time member-secretary. Board members, other than member secretary, have three years term which can be renewed twice.

Under the amended Act of 1988, the boards have been renamed as **Central Pollution Control Board and State Pollution Control Board** as they deal with both water and air pollution control. In order to meet the expenses as well as Control water pollution, a **Water (Prevention and Control of Pollution) Cess Act** was passed in 1977. It was amended in 1991.

Functions of Central Board :

- i) To advise Central Government and state boards about issues related to water pollution.
- ii) To provide technical assistance and guidance to state boards and industries.
- iii) To carry out and sponsor investigations and research related to water pollution.
- iv) To plan and organise training of persons in the field of water pollution.
- v) To utilise the services of mass media in educating public about causes, effects, prevention and control of water pollution.

- vi) To lay down standards for quality of water.
- vii) To establish and earmark laboratories, for sample analysis.
- viii) To look after pollution control in union territories, coordinate and oversee the work of state boards. In case of dispute the opinion of Central Board will prevail.

Functions of State Board :

- i) To advise State Government on matter concerning water pollution of industries likely to cause water or well pollution
- ii) To plan programme for prevention and control or abatement of water pollution.
- iii) To collect and disseminate information related to causes, prevention and control of water pollution.
- iv) To seek guidance and training of persons connected with prevention, control or abatement of water pollution from Central Board.
- v) To lay down and modify effluent standards and quality of receiving water.
- vi) To evolve economical and trade effluents.
- vii) To evolve methods for utilisation or disposal of treated sewage and effluents.

Powers :

1. Every industrial house must obtain consent on

- specific consent' application from State Board about the method of treatment and disposal of sewage or trade effluent, amount contents and place of discharge into a stream or well. A new consent is required for any addition or alternation. The board may impose conditions in the consent. The board reviews the conditions from time to time. It can order the firm to execute any work or install any machinery within a stipulated period.
2. It has the power to obtain information from individuals about removal or discharge of water, construction installation or operation of any disposal system in establishment where industry or trade is carried out.
 3. To take samples of any sewage or trade effluent, water from any stream and well.
 4. To enter any premises and inspect any plant, record, register, document and conduct search of a place if the authority has reason to believe that an offence under this Act has been committed. In residential areas the entry can be only at reasonable hours.
 5. In case of any emergency or pollution causing accident, the board can take remedial action in writing that may include restraining or prohibiting

discharge of pollutants into the open environment.

6. Normally the board makes an application to court after detecting an offence for giving injunctive relief restraining an establishment from causing pollution of water in stream or well.
7. Under the amended Act, the State Board can issue direction to any person, officer or authority about regulation, prohibition or closure of process, operation or industry. It can stop or regulate supply of water, electricity or any other service.
8. Statutory responsibility of involving prosecution lies with the Board. However, a private citizen can also invoke jurisdiction of court against erring industry after 60 days notice to board.

Appeal : Any person aggrieved by the orders passed by the State Board or in respect of consent application may prefer an appeal before appellate authority.

Punishment : In case of failure to provide information or comply with directions of the board (section 20), a person on conviction by Court is liable to be punished with imprisonment upto three months or and a fine of upto 10,000 rupees. In case the

failure continues a person can be punished with an additional fine of upto Rs. 5000 per day during which such failure continues. A person convicted of causing pollution due to nonobservance of standards is liable to be imprisoned for 1½ (one and a half year) years to six years or and a fine of upto ten thousand rupees. Continued failure invites an additional fine of 5000 rupees per day.

4:03:6 Air (Prevention and Control of Pollution) Act, 1981 (Amended in 1987)

The act was preceded by Bengal Smoke Nuisance Act 1905, Bombay Smoke Nuisance Act 1912 and Gujarat Smoke Nuisance Act 1963. Air (Prevention and Control of Pollution) Act was promulgated under article 253 of Constitution for prevention, control and abatement of air pollution by creating central and state boards. Central and State Pollution Control Boards created under Water Act 1974 also perform the functions of Central and State Boards for prevention and control of air pollution on the same terms and possessing the same powers. Central Board exercises powers and performs the functions of State Board for Union Territories either directly or by delegating powers. The board shall meet at least once

in every three months. The chairman can, however, convene a meeting of the board at such time as he thinks fit for any business of urgent nature. The board can also seek advice or assistance from any person.

Functions of Central Board

- i) To advise Central Government and State boards about improvement in the quality of air and issues related to air pollution.
- ii) To provide technical assistance and guidance to state boards and industries.
- iii) To plan and oversee execution of nation-wide programmes for prevention, control or abatement of air pollution.
- iv) To plan and organise training of persons in the field of air pollution.
- v) To utilise the service of mass media in educating public about causes, effects, prevention and control of air pollution
- vi) To lay down standards for quality air.
- vii) To establish or earmark laboratories for sample analysis.
- viii) To collect, compile and publish technical and statistical data relating to air pollution.

Functions of State Board

- i) To advise State Government on matters concerning air pollution, area to be declared air pollution control area and location of different industries likely to cause air pollution.
- ii) To plan programmes for prevention and control or abatement of air pollution
- (iii) To collect and disseminate information related to causes, prevention and control of air pollution
- (iv) To seek guidance and training of persons from Central Board.
- (v) To lay down standards for air quality in consultation with Central Board.
- (vi) To inspect air quality in air control area from time to time and take steps to reduce the pollution.

Powers

1. To recommend to State Government to declare any area or areas within the state as air pollution control area or areas. The whole of Punjab has been declared as air pollution control area.
2. On the recommendation of State Board, the State Government can direct any industry to use only approved fuel and approved appliances. It can prohibit the burning of any material.

3. Every industrial establishment has to move application of consent. After making inquiry and studying details, the State Board shall refuse or grant the consent, subject to such conditions like specifications of control equipment, its periodic checking, alterations and replacements, and specifications of chimney.
4. The board can restrict industrial operations in air pollution control areas.
5. It can withdraw consent given to any industry and impose new conditions.
6. The board can give instructions to authorities under Motor Vehicle Act regarding standards for automobile emissions.
7. Board officials have the power to enter any premises at all reasonable times and inspect compliance with requirements of the Act.
8. To call information from any industry with regard to pollutants emitted into atmosphere.
9. To take air or emission samples for analysis.
10. In case of any emergency due to accidents or unforeseen events causing pollution, in pollution control area the board can take remedial action in writing.
11. Statutory responsibility of invoking prosecution lies with the board. A private citizen can also

invoke jurisdiction of court against an erring industry after giving 60 days notice to board.

Under the Act, any person aggrieved of orders passed by the board can appeal before appellate authority. Punishment if found guilty, is the same as in modified water Act.

4:03:7 Motor Vehicle Act, 1938 (Amended in 1988)

1. The vehicle must have passed the test of VRDE Ahmadnagar, ARAI Pune or CMTTI Budni.
2. Each vehicle must have all the components of the standards laid down by Bureau of Indian Standards.
3. The vehicle must meet all the safety standards.
4. Manufacturer is to ensure that the vehicle does not cause pollution i.e., It is "pollution free".
5. The vehicle is fitted with tune up and catalytic converter.
6. Each vehicle must obtain "pollution under control" certificate every 3-6 months from dealers having approved test equipment.
7. New emission standards have been notified which are effective from April 1995 and April 2000.
8. All the future vehicles are to have engines based on unleaded petrol.

9. All the transport vehicles are to obtain fitness before registration, two years for new vehicle and every year subsequently.
10. Every vehicle has a maximum life, ten years for transport vehicles and 16 years for other vehicles.
11. The driver of a vehicle should be at least matriculate. (Minimum level of education required is now stands abolished).
12. The vehicle having hazardous goods should have prescribed labels. The packages containing such goods should also bear specific labels.
13. The drivers should have the knowledge of fundamentals in case the vehicle is involved in accident.

Advantages

- i) Motor Vehicle Act imposes condition on manufacturer to improve the design for reducing mechanical and electrical breakdown.
- ii) The owner is required to keep the vehicle fit. It will bring down air pollution especially in the urban areas where motor exhausts constitute 60-70% of total pollution.

Limitations

- a) Despite requirements of obtaining 'Pollution under control' certificates, very few vehicle owners do so.

- b) The facility for testing the degree of pollution is not available in small towns.
- c) Very old vehicles are seldom discarded.
- d) The trade of spare parts is flooded with substandard components resembling genuine ones.

4:04 Environment (Protection) Act, 1986

It is a comprehensive legislation for protection and improvement of environment and matters connected therein. According to Environment Protection Act, environment includes water, air and land; inter-relationships which exist among and between water, air and human beings and other living creatures, plants, microorganisms and property. Pollution has been defined as the presence in the environment of any solid, liquid or gaseous substance in such concentration so as to be injurious to environment.

The act does not have a separate official body at the state level. Central Government constitutes the authority for the purpose of exercising powers and performing functions under the Act. Officers are appointed by the Central Government. Central Government has delegated some powers to State Governments for issuance of directions, authorization

for hazardous waste disposal, submission of environmental audit statements and taking cognisance of such offences through state board.

Functions

1. Plan and execute nationwide programme for prevention, control or abatement of environmental pollution.
2. Coordinate activities of states, officers and other authorities under this and any other act related to objects of this Act.
3. Carry out and sponsor investigations and research.
4. Collect and disseminate information
5. Prepare manuals, codes and guides regarding prevention, control and abatement of environmental pollution.
6. Lay down standards for the quality of environment in its various aspects.
7. Lay down standards for emission or discharge of environmental pollutants from various sources. Different standards may be laid down for different sources. The latest modified standards are effective from 1.1.1994.
8. Demarcate areas in which certain industries/processes/operations shall not be carried out or shall be carried out subject to safeguards.

9. Lay down procedures and safeguards for the prevention of accidents which may cause environmental pollution, and remedial measures for such accidents.
10. Lay down procedures and safeguards for tackling of restricted, prohibited and hazardous substances.
11. Examine manufacturing processes, materials and substances as are likely to cause environmental pollution.
12. Establishment or recognition of environmental laboratories.

Powers

1. A person empowered by the Central Government shall have the right to enter any establishment at all reasonable times with such assistance as he considers necessary for (i) inspection and testing any equipment, industrial plant, manufacturing or other processes, record, register, document or any other material. (ii) Taking samples of air, water, soil or other substance from any premises for the purpose of analysis.
2. To give directions to authorities or person for taking steps for prevention, control and abatement of environmental pollution.

Penalties : Each failure of contravention under this Act is punishable with imprisonment for a term which may extend to five years or fine which may extend to one lakh rupees or both. In case of failure continues there would be additional fine of upto five thousand rupees per day after conviction. Imprisonment may extend to seven years, if the offence continues beyond a period of one year.

4:05 Efforts Taken in India to Ensure Environmental Protection

In India many laws have been legislated to protect and conserve nature and its resources, of which the following are worth mentioning.

Indian Fisheries Act, Smoke Nuisance Act, Forest Conservation Act, Mines and Minerals Act, The Factories Act, Water Act (Prevention and Control of Pollution), Wildlife Protection Act, Agricultural Pests and Diseases Act, Atomic Energy Act, Merchant Shipping Act (Harbour and Coastal Water Dumping of oil etc.), National Policy on Development and conservation of Mineral Resources, etc.

As a follow-up of Rio De Janerio summit of **"United Nations Conference on Environment and Development"** held in June 1992, Indian Government established the **"National Green**

Tribunal" under an Act passed by Parliament in June 2010.

With the establishment of **"Central Pollution Control Board"**, measures to protect the environment and keep the level of air and water pollution under check have been initiated. Following this, different States in India too followed suit and established "State Boards of Pollution".

In the beginning separate "Pollution Control Boards" were established to monitor and control air pollution and water pollution, both at the Centre and States. In 1988, as per the amendments brought in water and air Acts, these two kinds of boards, one for air and the other for water have been merged together and renamed as "Central Pollution Control Board" and "State Pollution Control Board" respectively.

The Central Pollution Control Board provides advices now and then to the Central Government, develops the necessary guidelines for regulation and bring required amendments as and when required. State Pollution Control Boards are entrusted with the responsibility of legislating the laws required for controlling and preventing environmental pollution and enforcing them in the respective State. It is the State Pollution Control Board that is empowered to

set the standards and norms for the acceptable levels of pollution of air, water and land and continuously monitor whether the factories and industries are functioning as per the norms prescribed.

4:06 National Green Tribunal

4:06:1 Origin of National Green Tribunal

During the **Rio De Janerio** summit of **United Nations Conference on Environment and Development** in June 1992, India vowed the participating nations to provide judicial and administrative remedies for the victims of the pollutants and other environmental damage. As a follow-up, India established the **National Green Tribunal (NGT)**, under an Act passed by the Parliament on 18.10.2010. This court can rightly be called a 'special' because India is the third country following Australia and New Zealand to have such a system. **Justice Lokeshwar Singh**, Pantia became its first Chirman. Currently it is chaired by **Justice Adash Kumar Goel**. The sanctioned strength of the tribunal is currently 10 expert members and 10 judicial members, although National Tribunal Act, 2010 allows for upto 20 each. The Chairman of the tribunal is the administrative head of the tribunal who also serves as the judicial member. Each bench of

the tribunal must consist of atleast one expert member and one judicial member. National Green Tribunal (NGT) has its headquarters located at New Delhi. Bhopal, Kolkata, Pune and Chennai are the other four places, from where the zonal branches of the tribunal function.

The Southern Branch of National Green Tribunal, the first permanent Zonal Branch became operational on 30.10.2012 with **Justice M. Chokalingam** as judicial member and **Mr. R. Nagendran** as expert member. It is presently located at Kalas Mahal, Kamarajar Salai, PWD Estate, Triplicane, Chennai - 600 005.

4:06:2 Role and Functions of NGT

The Tribunal has original jurisdiction of "substantial question relating to environment" (i.e. a community at large is affected, "damage to public health at broader level) and damage to environment due to specific activity" (such as pollution). However, no specific method is defined in law for determining "substantial" damage to environment, property or public health. There is restricted access to an individual, only if damage to environment is substantial. The Tribunal shall not be bounded by the procedure laid down under the code of Civil Procedure, 1908, but shall be guided by the principles

of natural justice. The powers of tribunal related to an award are equivalent to Civil Court and tribunal may transmit any order / award to civil court having local jurisdiction. The Bill specifies that an application for dispute related to environment can be filed within six months only when first time dispute arose (provided tribunal can accept application after six months if it is satisfied that appellant was prevented by sufficient cause from filing the application).

Also Tribunal is competent to hear cases for several Acts such as Forest (Conservation) Act, Biological Diversity Act, Environment (Protection) Act, Water & Air (Prevention & Control of Pollution) Acts etc. and also have appellate jurisdiction related to above Acts. The award should be implemented within a period of 30 days of award or order received by aggrieved party. The Bill says that decision taken by majority of members shall be binding and every order of Tribunal shall be final. However an appeal against order, award of Tribunal shall lie to Supreme Court provided appellant shall make an appeal within 90 days of commencement of award but Supreme Court can entertain appeal even after 90 days if appellant satisfies Supreme Court by giving sufficient reasons.

The tribunal is mandated to make and endeavour

for disposal of applications or appeals finally within 6 months of filing the same.

In short it could be said NGT has been established for effective and speedy disposal of cases relating to environmental protection and conservation of forests and other natural resources. As this system does not allow for frequent adjournment of cases, filing appeal after appeal against the orders passed, by knocking the doors of higher courts of judiciary, (against the order of the NGT, only the Supreme Court could be approached) speedy disposal of cases related to environment has been made possible.

4:07 Central Pollution Control Board

Central Pollution Control Board (**CPCB**) of India is a statutory organisation under the Ministry of Environment and Forests. It was established on 22 September, 1974 under Water (Prevention and Control of Pollution) Act, 1974. CPCB is also entrusted with the powers and functions under the Air (Prevention and Control of Pollution) Act, 1981. As per the provisions of Environment (Protection) Act, 1986, it provides technical advice to the Ministry of Environment and Forests and coordinates the activities of State Boards by offering technical assistance and guidance and resolves disputes among

them. It is an apex organisation in the country in the field of pollution control, as technical wing of Ministry of Environment and Forests.

4:07:1 Administrative Structure and Functions of CPCB

The Central Pollution Control Board is led by its Chairman, who is nominated by the Central Government. CPCB has its head office in New Delhi with 7 Zonal Offices and 5 Laboratories. The board has approximately 500 full time employees including engineers, scientists and environmental protection specialists.

Its important functions are :

- i) Conducting environmental assessment and research.
- ii) Maintaining national standards under a variety of environmental laws, in consultation with zonal offices, tribal and local governments.
- iii) Monitoring the water and air quality and maintaining respective quality data.
- iv) Working with industries and all levels of government in a wide variety of voluntary pollution - prevention programmes and energy - conservation efforts.

- v) Advising Governments of Union Territories about an industry or the pollution source causing water and air pollution.
- vi) Implementing the legislation relating to prevention and control of environmental pollutions along with the State Pollution Control Boards (SPCBs).

The CPCB performs various functions through the following nine major project heads.

1. Pollution assessment (Survey and Monitoring)
2. R & D (Research and Development) and laboratory management
3. Development of standards and guidelines for industry specific emissions and effluents standards.
4. Training
5. Information data base management and library
6. Pollution Control Technology
7. Pollution Control Enforcement
8. Mass awareness programmes and publications
9. Hazard waste management

4:08 State Pollution Control Board

All States in India have established their own boards to continuously monitor and control air, water

and noise pollution. State Pollution Control Boards (SPCBs) undertake appropriate measures to regulate industries in their respective states so as to ensure the norms and standards set by the CPCB with respect to industry specific emissions and effluents.

4:08:1 Tamilnadu Pollution Control Board (TNPCCB)

TNPCCB, established in 1982, functions with head office at Chennai, headed by Chairman, Member Secretary, 2 Additional Chief Environmental Engineers and 10 Joint Chief Environmental Engineers, 31 District Offices headed by District Environmental Engineers and 2 District Offices headed by Assistant Environmental Engineers. To assist analytical and scientific side, the Board has established 5 Advanced Environmental Laboratories and 10 District Environmental Laboratories.

4:08:2 Functions of TNPCCB

TNPCCB is implementing the Pollution Control Legislations and Rules and Notifications framed therein. In discharging the duties entrusted to it, the Board investigates, collects and disseminates data relating to water, air and land pollution, lays

down standards for sewage / trade effluent and emissions.

The field officers of the Board periodically inspect every industry under their jurisdiction to assess the adequacy of treatment measures provided to treat the effluent and gaseous emission. Board has classified the industries as per their pollution load for effective monitoring as follows :

Monitoring Periodicity for Inspection and Sampling

Classification of Industry	Inspection	Sampling
Red		
Large	Once in month	Once in a month
Medium	Once in Two months	Once in Two months
Small	Once in 3-4 months	Once in 3-4 months
Orange		
Large	Once in Two months	Once in Two months
Medium	Once in Three months	Once in Two months
Small	Once in 4-6 months	Once in 4-6 months
Green		
Large	Once in 3 months	Once in 3 months
Medium	Once in 6 months	Once in 6 months
Small	Once in a year	Once in a year

1. Granting Permission to Industries

TNPCB issues consent to new industries in two stages. Consent to establish is issued depending upon the suitability of the site before the industry takes up the construction activities and consent to operate is issued after installation of pollution control measures by the unit to satisfy the standards.

2. Penalising the Erring Industries

TNPCB issues show cause notices, takes legal actions and also issues directions for closure, stoppage of power supply, water supply etc. against erring industries/ agencies for non-compliance of pollution control legislations, conditions and standards.

3. Delegating Powers to Field Officers

TNPCB has delegated the power to the field officers for close monitoring. The Joint Chief Environmental Engineer / District Environmental Engineers / Assistant Environmental Engineers are empowered to issue / renew consent to orange and green category industries and to renew to red small category industries. They are also empowered to issue show cause notices to all the industries.

4. Assisting to Establish Common Effluent Plants

TNPCB plays a catalytic role in the implemen-

tation of Common Effluent Treatment Plants (CETPs) for small scale units like tanneries, textile dyeing units etc., located in clusters.

5. Taking Steps for Disposal of Hazardous Wastes

TNPCB is taking effective steps for the safe disposal of hazardous wastes and has completed the inventory of hazardous waste generating units and also identified sites for disposal of hazardous wastes.

6. Developing Environmental Awareness

TNPCB creates environmental awareness in the State through the Environmental Training Institute, Environmental Awareness Cell, Environmental Awareness Programmes, Environmental Pavilion constructed at the Periyar Science and Technology Centre (Chennai), NGO Cell, Publishing of Newsletters / Pamphlets on environmental issues, etc.

4:09 International NGOs and Environmental Protection

A Non-Government Organisation (NGO) is a social service organisation working towards a better society. It perseveres to bring a positive change by uniting people who share the common vision of a developed India and pay back to the society which helped us. It is run by volunteers who are ready to offer their free service for the betterment of the society. Even the

big non-government organisations operating at international level had a humble beginning, being founded by the efforts of one or two persons and later on grew and expanded by their dedicated work. NGO is a broad term, which includes charity organisations, advisory committees and various other professional organisations.

NGOs are involved in the whole spectrum of developmental activities from creating environmental awareness to undertake watershed development; from disaster management to sustainable livelihoods; from joint forest management with the government to giving inputs to government policies; They range from clubs which encourage nature camping agencies to organisations which undertake research and monitorings.

There are large number of NGOs in India and other countries that are exclusively working for environmental protection, conservation and awareness. The number of such NGOs in our country is more than any of the developing country. Increasingly, the government is viewing NGOs not only as agencies that will help them to implement their programmes, but also as partners in shaping policy and programmes.

NGOs are now playing an important role in

framing the environmental policy, mobilizing public support for environmental conservation and protecting the endangered species of forests and animals, creating awareness about the dangers in using driftnets in the commercial fishing industry.

The issues like future environmental protection, sustainable development and zero population growth are some of the major concerns of the environmental NGOs.

Environmental policies will achieve positive results only when they are addressed to local issues and solve the problems of local people. NGOs play here an important role in supplying inputs to the government about the needs and mind set of the local people due to their close contact with the community. Unless the needs of the people are identified and supported, sustainable development cannot be achieved. Policy makers and administrators should take care in selecting, financing and implementing projects which are aimed at promoting social welfare. They should not encourage the enterprises that promote private ownership and co-operation.

Some of the widely known international environmental organisations are 'Green Peace', 'Worldwide Fund For Nature', 'Conservation

International', 'Environmental Foundation for Africa', 'International Union for Conservation of Nature', 'Friends of the Earth', 'Earth First' etc.

Notable NGOs in India working in the field of environmental conservation are :

- i) Assam Science Society
- ii) Bombay Natural History Society
- iii) Centre for Science and Environment (CSE)
- iv) Centre for Environmental Education (CEE)
- v) CPR Environmental Education Centre (C.R. Ramaswami Aiyar Foundation)
- vi) Kerala Sastra Sahithya Parishad
- vii) Kalpavriksh
- viii) Narmada Bachao Andholan
- ix) Worldwide Fund For Nature
- x) Foundation for Revitalisation of Local Health Traditions (FRLHT)

Role of NGOs in the Protection of Environment in India

The NGOs constitute a worldwide network interacting with Governments, internal and intergovernmental organisations in shaping international environmental policies. They play the following roles :

- i) Creating awareness among the public on current environmental issues and solutions.
- ii) Facilitating the participation of various categories of stakeholders in the discussion on environmental issues.
- iii) conducting participatory rural appraisal.
- iv) Being involved in the protection of human rights to have a clean environment.
- v) Protecting the natural resources and ensuring the equitable use of resources.
- vi) Data generation on natural resources, time line history of villages.
- vii) Analysis and monitoring of environmental quality.
- viii) Transferring information through newsletters, brochures, articles, audiovisuals etc.
- ix) Organising seminars, lectures, and group discussions for promotion of environmental awareness.
- x) Helping the village administrative officials in the preparation, application and execution of projects on environmental protection.

4:09:1 Environmental Foundation of Africa (EFA)

EFA was founded in United Kingdom in 1992 as the 'Environmental Foundation for Sierra Leone'

(ENFOSAL) and started implementation in **Sierra Leone** (a Western Coastal Country of Africa) in 1993. During the height of the war years, the organisation shifted to adjoining the country **Liberia** and started working under the current name 'Environmental Foundation of Africa (EFA)'. Since 1999, it restarted operations in Sierra Leone. EFA-Sierra Leone (EFA-SL) is operating as the local partner to the international NGO EFA.

4:09:1:01 EFA's Important Work and Strategies

1. Protecting the environment of West African region and restoring the degraded lands and forests by making them greener.
2. Providing environmental education and raising awareness on environmental issues in all sectors of the populations, appreciation and understanding of environmental management and the impacts of environmental damage on lives and livelihoods.
3. Working with other partners, preventing the degradation of the environment and undertaking projects to conserve pristine forests and various species of fauna.
4. Promoting sustainable livelihoods for

environmental protection and conservation through community based initiatives such as eco-tourism, social forestry, organic farming, controlled grazing etc.

5. Providing technical assistance and advice in environmental issues.
6. Consolidating its operations in Gambia, Liberia and Sierra Leone while advocating region specific approach in addressing environmental problems in West Africa.

4:09:2 Worldwide Fund for Nature (WWF)

The World Wide Fund for Nature (WWF) is an international non-governmental organisation founded on 29th April 1961, working in the field of biodiversity conservation and reduction of human intervention in the environment. It was formerly named as the 'World Wildlife Fund', which still remains its official name in Canada and the United States of America. The organisation changed to its present name in 1986, to better reflect the scope of its activities. In India, its branch WWF-India was established as a Charitable Trust in 1969. This organisation is dedicated to the conservation of natural habitats and eco-systems in India.

With its network of State / Divisional and Field

Offices spread across the country to implement its programmes, WWF-India is the largest and one of the most experienced conservation organisations in the country. The Secretariat of this organisation functions from **New Delhi**. This organisation is a part of the WWF family with 27 independent national organisations. All the branches of WWF are coordinated by the WWF International, whose Head Quarters is located at **Gland in Switzerland**.

In order to suit India's specific ecological and socio-cultural situation WWF-India articulated its mission in 1987 as follows :

"The promotion of nature conservation and environmental protection as the basis for sustainable and equitable development".

4:09:2:01 Five Broad Programmes of The WWF-India Mission

1. Promoting India's ecological security; restoring the ecological balance.
2. Conserving biological diversity.
3. Ensuring sustainable use of the natural resource base.
4. Minimising pollution and Wasteful consumption.
5. Promoting sustainable life styles.

WWF-India implements its conservation programmes through Field Programmes, Public Policy, Education, Communications, NGO Networking and Resource mobilisation.

4:09:2:02 Key Environmental Programmes Undertaken by WWF-India

- i) The tiger conservation programme.
- ii) Freshwater and wetlands programme.
- iii) River dolphin conservation programme.
- iv) Wildlife trade monitoring
- v) Managing and monitoring forests, environmental law, information management and environmental education.

4:09:3 Conservation International (CI)

Conservation International (CI) is an American nonprofit environmental organisation established by **Spencer Beebe** and **Peter Seligmann** in 1987 with headquarters in **Arlington**, Virginia, U.S.A. Its goal is to protect nature as a source of food, fresh water, livelihoods and a stable climate.

4:09:3:01 Important Functions of CI

CI's work focuses on science, policy and partnership with businesses and communities. The organisation employs more than 1000 people and

works with more than 2000 partners in about 30 countries. CI has helped to establish 1000 protected areas across 77 countries and protected more than 612 million hectares of land, marine and coastal areas.

The aim of CI is to analyse the problems most dangerous or harmful to nature and build a foundation dedicated to solve these issues on a global scale. This organisation :

- i) detects the problems most threatening to nature
- ii) prevents the industry side of the world from being detrimental to nature
- iii) ensures the knowledge the institution has acquired over its first 25 years is shared with governments and in doing so, establishes policies within these countries that serve as a great benefit to people and nature.

In 1989, CI formally committed to protection of biodiversity hotspots, ultimately identifying 34 such hotspots around the world and contributing to their protection. In 2008, this organisation's leadership believed that CI's focus on biodiversity conservation was inadequate to protect nature and those who depended on it. As a result CI's work focuses on (i) Climate change (ii) fresh water security (iii) health (iv) food security (v) biodiversity and

(vi) cultural services. This organisation's present slogan is

"Make conservation a cornerstone of economic development to benefit everyone, everywhere"

At present, the foundation of CI's work is "Science, partnership and field demonstration". The organisation has scientists, policy workers and other conservationists on the ground in more than two dozen countries, on five continents. It also relies heavily on hundreds of local partners.

4:09:4 Green-Peace

'Green Peace' is an environment-friendly international non-profit organisation founded in America in 1971. At present it is functioning with Amsterdam of Netherland as its head quarters. It aims at promoting environmental awareness. It is an independent, campaigning organisation, addressing the environmental abuse through direct, non-violent confrontations with governments and business companies. It exposes the global environmental problems and provides solutions for a healthy environment. It does not accept funds from governments or from the commercial firms. It depends mainly on the voluntary contribution, of individuals and grants from charitable foundations.

Green-peace was founded in 1971 to oppose nuclear testing in **Alaska**. French intelligence agents blew up **Rainbow Warrior**, a Green-peace ship scheduled to protest French nuclear weapons tests, in **Auckland Harbour, Newzeland** on 10th July, 1985. The resulting scandal caused the resignation of the French defense minister and the firing of the head of the French Intelligence Service.

4:09:4:01 Mission of Green-peace

Green-peace focuses on the most crucial worldwide threats to our planet's biodiversity and environment. It campaigns to :

1. Stop Climate-Change

The extensive use of oil, gas, fuel, and other energy resources leads to climatic changes which results in global warming. In order to stop climate changes, Green-peace is campaigning on various fronts. It has been researching to stop climate change and to promote clean energy solutions.

2. Protect Ancient Forests

Many forests of the world are in crisis. The plants and animals are facing the threat of extinction. People living in forests and depending on them for their livelihood are also under threat. Green-peace takes

up the responsibility to save the forests and provides solutions for the same.

3. Save the Oceans

Green-peace's save the oceans campaign currently focuses on four major threats to the world's oceans; overfishing, pirate fishing, whaling, and intensive shrimp aquaculture.

4. Stop Whaling

Commercial Whaling has resulted in the decline of the world's whale population. In order to stop commercial whaling, Green-peace is working on many fronts. Through political work, public outreach and by adopting nonviolent direct action against the whalers at sea. Green-peace is fighting against commercial whaling.

5. Say 'No' to Genetic Engineering

Genetic engineering enables creation of plants, animals and micro-organisms through the manipulation of genes. The organisms, which are produced through genetic engineering when interbred with the natural organisms lead to new environments, which are uncontrolled, causing "**genetic pollution**". Therefore Green-peace opposes the release of genetically doctored varieties of plants and animals.

6. Stop the Nuclear Threat

Green-peace campaigns against the use of nuclear power as its use has never been peaceful. It leads to accidents, deaths and disasters. Radiation released into the environment through nuclear tests has led to contamination of soil, air, rivers and oceans, causing cancer and other diseases. It is worth recalling here that Green-peace campaigned against Kudangulam Atomic Power Project in India under the leadership of Mr. Udayakumar.

7. Eliminate Toxic Chemicals

Green-peace also campaigns against toxic chemicals, as they prove to be a global threat to the health and environment. It urges governments to ban chemicals like DDT, Malatheaon etc.

8. Encourage Sustainable Trade

Green-peace opposes the current form of globalisation that is increasing corporate power. It demands that the World Trade Organisation (WTO) adopt a policy of trade, which works for all and that preserves and restores environment. Governments must work toward achieving sustainable development, which means integrating three things, environmental, social and economic priorities.

4:09:4:02 Major Achievements of 'Green - peace'

1. A ban on toxic waste exports to less developed countries.
2. A moratorium on commercial whaling.
3. A United Nations Convention providing for better management of World fisheries.
4. A Southern Ocean Whale Sanctuary.
5. A 50 year moratorium on mineral exploitation in Antarctica.
6. Ban on the dumping of the sea of radio active and industrial waste and disused oil installations.
7. An end to high-sea, largescale driftnet fishing.
8. A ban on all nuclear weapons testing obtained as a result of Green-peace's first ever campaign.

4:09:5 International Union for Conservation of Nature (IUCN)

IUCN was established in France in October 1948 and its headquarters at present is located in **Gland, Switzerland**. It was previously called the **International Union for Protection of Nature** (1948-1956), **Worldwide Fund for Nature** (1956-1990) and the **World Conservation Union** (1990-2008). Its full legal name is **International Union**

for Conservation of Nature and Natural Resources. It is shortly referred to as IUCN at present.

4:9:5:01 Functions of IUCN

IUCN is working in the field of nature conservation and sustainable use of natural resources. It is involved in data gathering and analysis, research, field projects, advocacy and lobbying and education.

4:9:5:02 Aims of IUCN

IUCN's mission is to "influence, encourage and assist societies throughout the world to conserve nature and ensure that any use of natural resources is equitable and technologically sustainable.

Over the past decades, IUCN had widened its focus beyond conservation ecology and now incorporates issues related to gender equality, poverty alleviation and sustainable business in its projects.

4:9:5:03 Nature and Activities of IUCN

Unlike other NGOs, IUCN does not itself aim to mobilize the public in support of nature conservation. It tries to influence the actions of governments, business and other stakeholders by providing information and advice, and through lobbying and

partnerships. The organisation is best known to the wider public for compiling and publishing the **ICUN Red List of Threatened Species**, which assesses the conservation status of species worldwide.

4:9:5:04 Organisational Structure of IUCN

IUCN has a membership of over 1400 governmental and non-governmental organisations. Some 16,000 scientists and experts participate in the work of IUCN Commissions on voluntary basis. Its organisation structure has three components : the member organisations, the six scientific commissions and the secretariat. The members can organise themselves in national or regional committees to promote cooperation. In 2014, there were 56 National Committees and 7 Regional Committees. It employs 1000 full time staff in more than 60 countries. From 1991 it publishes the journal '**Caring for the Earth**' jointly with United Nations Environment Programme (UNEP) and Worldwide Fund for Nature (WWF).

IUCN works on the basis of four year programmes determined by the General Conference of the membership. The three thrust areas identified by the work programme are :

- i) Valuing and conserving nature

- ii) Effective and equitable governance of nature's use
- iii) Deploying nature-based solutions to global challenges in climate, food and development.

Conclusion

In this unit important environmental issues like 'Global Warming', 'Green House Effect', 'Ozone Layer Depletion', 'Acid rain', 'Extinction of Flora and Fauna' etc., the policies and measures adopted by the governments both at the Centre and States, Various laws and rules legislated by the governments have been discussed elaborately. Further the role of NGOs in the protection of environment, especially Environment Foundation of Africa, Worldwide Fund for Nature, Conservation International, Green Peace and International Union for Conservation of Nature have also been discussed in detail.

Review Questions :

1. What is 'Global Warming'? Explain Green House Effect leads to global warming. B. [Ans: 4:01:1 + 4:01:2]
2. Explain the causes and effects of climate change. (B) [Ans: 4:01:3]

3. What are the causes for ozone layer depletion. (C) [Ans: 4:01:4:01]
4. What is 'Acid Rain'? How does it occur? (B) [Ans: 4:01:6]
5. Discuss the causes and consequences of the extinction of flora and fauna and suggest measures to control it. (A) [Ans: 4:01:7 + 4:01:7:01 + 4:01:7:02]
6. What do you mean by conservation of Environment? What is the need for it? (C) [Ans: 4:02:1 + 4:02:2]
7. List the environmental conservation measures taken in India. (B) [Ans: 4:02:4]
8. Discuss the different constitutional amendments made to safeguard the Indian environment (A) [Ans: 4:03 + 4:03:1 to the end of 4:03:7]
9. Briefly explain the following :
 - i) Factories Act (C) [Ans: 4:03:1]
 - ii) Environment Conservation Act (B) [Ans: 4:03:2]
 - iii) Forest (Conservation) Act (B) [Ans: 4:03:3]
 - iv) Wild life (Protection) Act (B) [Ans: 4:03:4]
 - v) Water (Prevention and Control of Pollution) Act (B) [Ans: I Paragraph of 4:03:5 + Functions of Central Board + Functions of State Board + their Powers]
 - vi) Air (Prevention and Control of Pollution) Act. (B) [Ans: 4:03:6]
 - vii) Motor Vehicle Act (B) [Ans: 4:03:7]